

Conjecture 00000002181 is false: parity has sensitivity n and spectral norm 1

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1 The conjecture

Conjecture 00000002181 (English text, verbatim): “Definition: The Gotsman–Linial conjecture relates the sensitivity of Boolean functions to their spectral norm. Conjecture: Sensitivity is always at most $\sqrt{2} \cdot \log n$ times the spectral norm; and the bound is controlled by the pointer maxima of real hypercubical faces.” The Chinese text says the same (sensitivity never exceeds the spectral norm times a $\sqrt{2} \log n$ factor; and the bound is controlled by pointer maxima of real hypercube faces).

Reading. The conjecture is a conjunction. Its first clause says: for every Boolean function f on n variables,

$$s(f) \leq \sqrt{2} \log n \cdot \|\hat{f}\|_1. \quad (*)$$

Here n is the number of input variables. We refute (*), which refutes the conjunction; the second clause (“pointer maxima of real hypercubical faces”) is not used. We refute (*) under each of the following choices, and do not claim more:

- sensitivity read as maximum sensitivity, as sensitivity at a point, or as average sensitivity (all equal n for the parity witness);
- spectral norm $\|\hat{f}\|_1 = \sum_S |\hat{f}(S)|$ of the ± 1 encoding $(-1)^f$ or of the 0/1 encoding of f ;
- $\log$ the natural logarithm or $\log_2$; in fact $\sqrt{2} \log n$ may be replaced by $C \log n$ for any constant C , which covers every base $b > 1$ since $\log_b n = \log n / \log b$;
- restricting (*) to $n \geq 2$ (to avoid the degenerate value $\log 1 = 0$), or to linear threshold functions.

2 Definitions and conventions

Inputs are $x \in \{\text{false}, \text{true}\}^n$ (Lean: `Fin n -> Bool`). The sign of a bit is $\text{sgn}(\text{false}) = 1$, $\text{sgn}(\text{true}) = -1$. The point $x^{\oplus i}$ is x with bit i flipped.

Definition 1. $s(f, x) = \#\{i : f(x^{\oplus i}) \neq f(x)\}$; the sensitivity is $s(f) = \max_x s(f, x)$; the average sensitivity is $2^{-n} \sum_x s(f, x)$. For $S \subseteq [n]$ the Walsh character is $\chi_S(x) = \prod_{i \in S} \text{sgn}(x_i)$. For $g : \{\text{false}, \text{true}\}^n \rightarrow \mathbb{R}$ the (normalised) Fourier coefficient is $\hat{g}(S) = 2^{-n} \sum_x g(x) \chi_S(x)$, and the spectral norm is $\|\hat{g}\|_1 = \sum_S |\hat{g}(S)|$. The ± 1 encoding of f is $x \mapsto \text{sgn}(f(x))$, the 0/1 encoding is $x \mapsto [f(x) = \text{true}]$. Parity is $\text{PAR}_n(x) = \text{true}$ iff $\#\{i : x_i = \text{true}\}$ is odd; $\text{AND}_n(x) = \text{true}$ iff all

$x_i = \text{true}$. A linear threshold function is one of the form $f(x) = [\theta \leq \sum_i w_i x_i]$ with real w_i, θ (bits read as 0/1).

3 Proof

Lemma 2 (`sum_prod_bits`). $\sum_x \prod_i h_i(x_i) = \prod_i (h_i(\text{true}) + h_i(\text{false}))$.

This is distributivity (`Mathlib Fintype.prod_sum`).

Lemma 3 (`sgn_parity, parity_flipBit`). $\text{sgn}(\text{PAR}_n(x)) = \prod_i \text{sgn}(x_i)$, hence flipping any bit changes $\text{PAR}_n(x)$.

Proof. The product equals $(-1)^{\#\{i:x_i\}}$. Flipping bit i negates exactly one factor. $\square$

Theorem 4 (`parity_sensAt, parity_sensitivity, parity_avgSensitivity`). $s(\text{PAR}_n, x) = n$ for every x ; so $s(\text{PAR}_n) = n$ and the average sensitivity is n .

Theorem 5 (`fourierCoeff_pm_parity, spectralNorm_pm_parity, spectralNorm_zo_parity`). The ± 1 parity has $\hat{g}(S) = [S = [n]]$, so $\|\hat{g}\|_1 = 1$. For $n \geq 1$ the 0/1 parity has $\hat{g}(S) = \frac{1}{2}[S = \emptyset] - \frac{1}{2}[S = [n]]$, so $\|\hat{g}\|_1 = 1$.

Proof. $\text{sgn}(\text{PAR}_n(x))\chi_S(x) = \prod_i h_i(x_i)$ with $h_i(b) = \text{sgn}(b)^2 = 1$ if $i \in S$ and $h_i(b) = \text{sgn}(b)$ otherwise. By the first lemma the sum over x is $\prod_i (h_i(\text{true}) + h_i(\text{false}))$, which is 2^n if $S = [n]$ and 0 otherwise. Likewise $\sum_x \chi_S(x) = 2^n[S = \emptyset]$. The 0/1 encoding equals $(1 - \text{sgn} \circ f)/2$, which gives its coefficients; $\emptyset \neq [n]$ when $n \geq 1$. $\square$

Lemma 6 (`sqrt_two_mul_log_lt, sqrt_two_mul_logb_lt, const_mul_log_lt`). For every integer $n \geq 1$: $\sqrt{2} \ln n < n$ and $\sqrt{2} \log_2 n < n$. For every real C there is N with $C \ln n < n$ for all $n \geq N$.

Proof. From $\ln y \leq y - 1$ with $y = n/e$ we get $\ln n \leq n/e$. Since $\sqrt{2} < 3/2 < e$, the first claim follows. For $\log_2$: $e \ln 2 > 2.718 \cdot 0.693 > 3/2 > \sqrt{2}$, so $\sqrt{2} \ln n < e \ln 2 \ln n \leq n \ln 2$ when $n \geq 2$, and $n = 1$ is trivial. For the last claim, `Mathlib's Real.log_le_rpow_div` with $\varepsilon = 1/2$ gives $\ln n \leq 2\sqrt{n}$, and $|C| \cdot 2\sqrt{n} < n$ once $\sqrt{n} > 2|C| + 1$; we take $N = \lceil (2|C| + 1)^2 \rceil + 1$. $\square$

Theorem 7 (`parity_violates, conjecture_2181_false, no_log_bound`). For every $n \geq 1$, $\sqrt{2} L(n) \|\hat{g}\|_1 < s(\text{PAR}_n)$ for $L \in \{\ln, \log_2\}$ and g either encoding of PAR_n . Hence (*), even restricted to $n \geq 2$, is false for both logarithms and both encodings (witness $n = 2$: $s = 2$, $\|\hat{g}\|_1 = 1$, $\sqrt{2} \log_2 2 \approx 1.414$, $\sqrt{2} \ln 2 \approx 0.980$). Moreover, for every constant C , $C \ln n \|\hat{g}\|_1 < s(\text{PAR}_n)$ for all large n .

Theorem 8 (`andFun_isLTF, andFun_sensitivity, spectralNorm_zo_and, spectralNorm_pm_and_le, and_no_log_bound`). AND_n is a linear threshold function (weights 1, threshold n), $s(\text{AND}_n) = n$ (at the all-true input), its 0/1 encoding has spectral norm 1 (every coefficient has absolute value 2^{-n}) and its ± 1 encoding has spectral norm at most 3 (since $\text{sgn} \circ f = 1 - 2 \cdot [f]$). So for every C , $C \ln n \|\hat{g}\|_1 < s(\text{AND}_n)$ for all large n : the bound fails even on linear threshold functions.

4 Lean formalisation and axiom audit

File `lean/Conjecture2181/Basic.lean` (Lean 4, Mathlib v4.33.1, only `import Mathlib`). Main statement:

```
def FirstClause (L : R -> R)
  (enc : forall {n}, BoolFun n -> (Fin n -> Bool) -> R) :=
  forall n, 2 <= n -> forall f : BoolFun n,
    (sensitivity f : R) <= sqrt 2 * L n * spectralNorm (enc f)
theorem conjecture_2181_false :
  ~ FirstClause Real.log pmEnc /\ ~ FirstClause (Real.logb 2) pmEnc /\
  ~ FirstClause Real.log zoEnc /\ ~ FirstClause (Real.logb 2) zoEnc
```

(written here in ASCII; $R = \mathbb{R}$). Also `parity_violates`, `no_log_bound`, `and_no_log_bound`, `andFun_isLTF`, `parity_avgSensitivity`. `lake build` succeeds, and `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound` for each of these theorems; there is no `sorry` and no `native_decide`.